MIL-S-12653B(Ord)
16 May 1961
SUPERSEDING
MIL-S-12653A(Ord)
24 June 1954

MILITARY SPECIFICATION

SIGHT, LEAD COMPUTING, AUTOMATIC: M38

1. SCOPE

1.1 This specification covers one type of computing sight for use on Gun, Self-Propelled, Twin 40MM: M42 and M42A1, hereinafter referred to as "computing sight".

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids form a part of this Specification to the extent specified herein.

SPECIFICATION

Military

MIL-O-13830	Optical Components for Fire Control Instruments; General specification Governing the Manufacture and Inspection of
MIL-F-13926	Fire Control Materiel; General Specification Governing the Manufacture and Inspection of
MIL-S-14487	Sight, Computing, M38, Packaging of
MIL-I-45208	Inspection Requirements; General Specification for

DRAWINGS

Ordnance Corps

F7665137	Sight, Computing, M38
F822715	Test Fixture

FSC 1220

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STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-130	Identification Marking of U.S. Military Property

(Copies of Specifications, standards, and drawings, required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Production model.- At the beginning of production one computing sights respresentative of production, shall be made available to the Government to determine if the computing sight meets all the requirements of this specification.

3.2 Construction.- Construction of the computing sight shall be in accordance with Drawing F7665137 and all drawings pertaining thereto.

3.3 Materials.- Materials shall be in accordance with drawings, material specifications, and general specifications forming a part of this specification.

3.4 General specification.- The contractor shall be responsible for compliance with the following requirements of Specifications MIL-O-13830 and MIL-F-13926.

3.4.1 MIL-O-13830

(a) Cleanliness

3.4.2 MIL-F-13926

(a) Order of precedence
(b) Protective finished surfaces
(c) Special accuracy
(d) Finish, painting and protective
(e) Workmanship

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3.5 Adjustment.

3.5.1 Elevation linkage.- The parallelogram linkage support (with pads for quadrant on Body,F7665140) shall repeat the simulated gun elevation on the test stand within plus or minus 1 roil, after it has been adjusted in accordance with 4.7.

3.5.2 Boresighting.- Telescope adjustments shall be set to within plus or minus 1 mil of alinement with the boresight mark on the test target.

3.5.3 Superelevation.- Superelevation shall be set to 9 mils plus or minus 1 mil at zero elevation.

3.5.4 Bail brake.- Bail brake adjustment shall be in accordance with 4.7(i).

3.6 Environmental.

3.6.1 Vibration.- The computing sight shall be subjected to a simple harmonic motion having an amplitude of 0.03-inch (0.06-inch maximum excursion). The frequency shall be varied uniformly between the limits of 10 and 55 cycles per second and return to 10 cycles and shall be traversed in approximately 1 minute. This motion shall be applied for a period of 10 minutes in each of the 3 mutually perpendicular directions (total of 30 minutes). Subsequent to vibration the computing sight shall not be damaged or adversely affected and shall meet all requirements of this specification.

3.6.2 Shock. The computing sight shall show no evidence of damage and shall meet the performance requirements of this specification after exposure to shock impacts of 20 gravitational units (g's) applied 3 times in each of its 3 mutually perpendicular planes.

3.6.3 Humidity.- The computing sight shall show no evidence of damage and meet the performance requirements of this specification imediately after being subjected to 1 cycle of the humidity cycle shown in Figure 1.

3.6.4 Rain.- The computing sight shall meet the performance requirements of this specification immediately after being subjected to a 2 hour simulated rainfall of 4 plus 1 minus 0 nches-per-hour with Dust Cover Assembly F7666267 installed, and 1/4 plus or minus 1/8 inch-per-hour with dust cover assembly removed.

3.6.5 Operating temperature.- The Computing Sight shall meet the performance requirements of this specification while thermally stabilized at ambient temperatures of plus 150 and minus 40 degrees, Fahrenheit.

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3.6.6 Storage temperature. - The computing sight shall show no evidence of physical failure and meet the performance requirements of this specification at standard ambient temperature (plus 60 to 90 degrees, Fahrenheit) after having been thermally stabilized at the ambient temperatures of plus 160 and minus 80 degrees, Fahrenheit.

3.7 Performance.

3.7.1 Limits of operation.- The computing sight shall operate satisfactorily between the limits of minus 6 degrees and plus 90 degrees simulated gun elevation on the test fixture.

3.7.2 Problem accuracy.- The computing sight shall compute deflection to an accuracy of 5 mils average error radially (see 4.8.2), except at zero speed with zero elevation and plus 1600 mils angle of approach, where the accuracy shall not exceed 2 roils average error radially. At least 80 percent of the average errors shall fall within the required tolerance and no reading at any check point shall exceed an error of 10 mils radially. The “backlash figure of merit” (see 4.8.2) shall not exceed 4.5 mils.

3.7.3 Backlash.

3.7.3.1 Vertical.- Vertical backlash measured at the telescope shall not exceed 2 mils at 700 miles per hour, zero climb and dive, and plus 1600 mils angle approach, when tested in accordance with 4.8.3.1.

3.7.3.2 Azimuth.- Azimuth backlash measured at the telescope shall not exceed 4 mils at zero miles per hour,zero climb and dive, and zero mils angle of approach and shall not exceed 6 roils at 700 miles per hour, zero climb and dive, and plus 1600 mils angle of approach, when tested in accordance with 4.8.3.2.

3.7.3.3 Slide.- The backlash measured at the base of Slide C7675873 shall not exceed 0.015 inch, total indicator reading, at zero speed, zero climb and dive, and zero angle of approach, when a force of 5 pounds is alternately applied in opposite directions at the end of the slide.

3.7.3.4 Handwheel.- Backlash between the course handwheel and the azimuth indicating scale on the computer head shall not exceed 5 degrees measured at the course handwheel.

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3.7.4 Torque. Tests shall be conducted at temperatures between 60 and 90 degrees Fahrenheit.

3.7.4.1 Shaft.- With the computing sight and fixture set at zero elevation, the torque required at Shaft B7665145 to operate the sight in clockwise and counterclockwise directions shall not exceed 25 ounce inches when measured at angle of approach settings of zero mils, plus 1600 mils, plus 3200 mils, and minus 1600 mils, with zero speed and zero inclunation settings.

3.7.4.2 Handwheel.- The torque required at Handwheel Assembly D7666207 to rotate the computer head shall be between 18 and 25 pound inches.

3.7.5 Tape tension.- Tape tension shall provide a torque between 50 and 80 pound inches at the telescope, when measured by applying torque in a counterclockwise direction *only,* for 5 degrees movement from a reference point.

3.7.6 Air speed knob.- With the angle of approach set at plus 800 mils, dive and climb angle set at 40 degrees dive, and gun elevation set at zero, the torque required to rotate the air speed knob shall be between 15 and 45 pound-inches, when measured in accordance with 4.7(j).

3.7.7 Identification marking.- Identification marking unless otherwise specified shall be in accordance with Standard MIL-STD-130.

4. QUALITY ASSURANCE PROVISIONS

4.1 General provisions.- The quality assurance provisions of referenced documents to the estent specified herein, together with provisions of this section, are applicable to inspection performed by the contractor and for Government verification.

4.1.1 The contractor shall provide an inspection system in accordance with Specification MIL-I-45208.

4.2 Production model.- Unless otherwise specified in the contract, one computing sight shall be selected by the Government inspector as soon as practicable after the start of production, and be inspected by the contractor under the surveillance of the Government. This inspection shall be conducted at the contractor's facility, or Government approved laboratory, using test equipment authorized for use on the contract. The ample shall be inspected for all requirements of this specification by utilizing examination and testing methods prescribed herein.

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4.2.1 Production model failure.- Should the production model fail to pass the prescribed tests, the contractor shall cease those operations which are causing the rejection and shall make the necessary corrections. When corrections in the process have been made, a new production model shall be examined and tested by the contractor under the surveillance of the Government. Acceptance *of* production lots will not commence until an acceptable production model is submitted and corrective actions have been taken to eliminate deficiencies.

4.3 Lot formation.- Inspection lot sizes and lot formation shall be in accordance with Standard MIL-STD-105.

4.3.1 Defective units and lots shall be processed as specified in Standard MIL-STD-105.

4.4 Inspection of components.- All positional and functional characteristics shown on the drawings contained in the contract, which indicate precision requirements and have a bearing on functional accuracy and performance of the computing sight shall be inspected in accordance with the written inspection system required by Specification MIL-I-45208. These characteristics are defined as drawing requirements affecting concentricity, squareness, parallelism, micro-finishes 64 rms or better, threads, identification marking, and performance requirements specified in notes on the drawings and in separate performance specifications referenced on the drawings.

4.4.1 Sampling inspection.- Sampling inspection of components may be used in lieu of 100 percent inspection, and shall be in accordance with-Table I.

TABLE I - SAMPLING PLAN

Lot Size Inclusive	Sample Size	Accept Lot	Reject Lot
0 - 25	Use 100% Inspection		
26 - 35	20	0	1
36 - 50	25	0	1
51 - 150	30	0	1
151 - 300	30	1	2
301 - over	60	2	3

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4.4.2 Disposition of non-conforming product.- Acceptance and rejection numbers are expressed in terms of defective units. One or more defects on any unit shall constitute a defective unit. Disposition of non-conforming components shall be in accordance with Standard MIL-STD-105.

4.5 Material.- Control of material shall be in accordance with test data requirements of Specification MIL-I-45208.

4.6 Inspection equipment.- Supply, calibration, maintenance, and use of inspection equipment shall be in accordance with Specification MIL-I-45208.

4.6.1 Accuracy requirements.- Inspection equipment which incorporates features for measuring values or sizes established by product requirements shall be capable of measuring those values within an acouracy of ten percent of the specified tolerance, except as otherwise specified. In case where the ten percent accuracy imposes impractical or very difficult design and fabrication problems, or results in a test equipment cost factor which is exorbitant in relation to the cost and application of the product to be inspected; the Government shall be expeditiously advised in order that necessary appropriate action can be taken.

4.6.2 Accuracy requirements for special testing devices.

4.6.2.1 Device to test computing sight accuracy.- The device utilized to final test the computing sight shall be correlated to Test Fixture 8227175, Drawing F822715. It shall simulate the gun breech casing with respect to mounting, and shall provide a means for introducing elevation into the sight between the limits of minus 6 degrees and plus 90 degrees. The centerline of the test fixture shall be level and perpendicular within 0.5 mil to a like centerline on the test target. The fixture also shall contain a power device capable of driving the computer head in azimuth, in both a clockwise and counterclockwise direction at 5 plus or minus 1 degree per second and 45 plus or minus 2 degrees per second.

4.6.2.2 Test target.- The test target shall accommodate maximum sight deflection for a target distance of not less than 18 feet. Layout of check points shall be in accordance with “Test Target Layout Parameters” shown in Figure 2 of this specification.

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4.6.2.3 Telescope.- Since the reflex sight has a significant inherent parallax when used in conjunction with a close test target, a straight tube telescope of the M7 type shall be used in making the necessary tests and ajustments. An adapter shall be used for mounting the telescope so that the heiht of the optical axis above the center of rotation shall be 6.186 inches.

4.7 Test adjustments.- With the computing sight set upon the test fixture, the following adjustments and checks shall be made:

(a) With the test fixture at zero elevation, set angle of approach at zero mils dive and climb bail in a vertical position, and speed knobs against the zero stop. Adjust Rod C7665174 to obtain a zero reading wioth a zeroed gunner's quandrant placed on the machined pads of Body F7665140. Place ther gunners quadrant on the machined pads of Body Assembly D7665230 and adjust Stem B764832 of the mall joint assembly at the bottom of the computer head to a zero reading on the quadrant.

(b) Check tape tension. It shall meet the requirements of 3.7.5.

(c) With the test fixture at zero elevation, set angle of approach at approximately plus 1600 mils. Maintain position of dive and climb bail and air speed knobs as set in (a) above. Boresight the telescope to a reference point on the target. Power rotate the computer head at 5 plus or minus 1 degree per second and 45 plus or minus 2 degrees per second in both clockwise and counterclockwise directions. Movement of the telescope vertical reticle line shall not exceed an excursion of 4 mils when viewed on the target for each of the rates indicated when checked in each direction of rotation. If excursion of more than 4 mils is visually apparent, it may be necessary to make adjustments to the stop screws in the air speed knob Assembly C7666191.

(d) With the test fixture at 1600 mils elevation position, set the angle of approach at zero. Adjust the angle of approach until the line of sight through the telescope remains on the horizontal reference line within 0.5 mil while the speed knob is rotated from 0 to 700 miles per hour. Set the angle of approach index to coincide with the zero graduation on the computer head.

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(e) With the test fixture at zero elevation, set angle of approach to plus 1600 mils. Rotate the speed knob 0 to 700 miles per hour, while adjusting the position of the computer head in a horizontal plane by means of the computer head pivot Shaft B7666436 until the line of sight of the telescope remains on the target reference line within 0.5 mil.

(f) With the test fixture at zero elevation, set angle of approach at 3200 mils. Adjust the climb and dive bail until the line of sight of the telescope follows a horizontal reference line within 1 roll, when the speed knob is rotated from 0 to 700 miles per hour. Set the ball indices to zero. Repeat at zero mils angle of approach.

(g) With the test fixture at zero elevation, set angle of approach at plus 1600 mils and the speed knob at 700 miles per hour. Move the ball through the full length of travel. The line of sight of the telescope shall remain on a vertical reference line within 1 mil.

(h) Introduce superelevation into the sight as follows: With the test fixture at zero elevation, set the anglew of approach to plus 1600 mils. Set speed knob at boresight mark (BS) and set in 50° dive angle. Place a zeroed gunners quadrant on the machined pads on Body Assembly D7665230 and obtain a zero reading by adjusting Stem B7648328 at the bottom of the computer head. The telescope will now be alined on the boresight mark. Set the speed knob at zero speed and dive ball to zero. Superelevation is now set into the sight. Observe the line of sight; it shall meet the requirements or 3.5.3B. Introduce a 50° dive angle and set speed knob to boresight mark. This will remove superelevation. The line of sight should now be on the boresight mark of the test target.

(i) Bail brake adjustment is to be set so that when rotating the speed knob slowly through the entire range, so that individual speed clicks may be felt and counted, the bail shall not move more than the width of a graduation line on the dive and climb scale. The force required to move the bail, measured with a spring scale applied to the bail at the course arrow, shall not exceed 40 pounds for any position, of the ball between 65° dive and 55° climb. The above bail efforts are to be determined with the speed set at 600 mph, angle of approach set at plus 800 mils, and gun elevation set at zero.

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(j) Air speed knob torque shall be measured at 200 mph, 400 mph, and 600 mph with the computer course, dive, and elevation set in accordance with 3.7.6. Torque measurements shall be made by repeated 20 mph reversals of the air speed knob. The first two reversals for a given speed setting shall not be measured.

4.8 Performance tests.

4.8.1 Classification of defects,- The classification of defects in Tables II and III shall constitute the minimum inspection requirements to be performed prior to acceptance or rejection by lots. The Government reserves tghe right to inspect for any or all applicable requirements and to reject individual non-conforming items.

4.8.1.1 Disposition of rejected lots.- Rejected lots shall be screened for all defective characteristics. After corrections have been made and the Government inspector infomed of the corrective action taken, the lots shall be resubmitted for acceptance as outlined in Standard MIL-STD-105.

TABLE II CLASSIFICATION OF DEFECTS

Use Inspection Level 7, and Sampling Plan Table IV-A of Standard MIL-STD-105

CRITICAL: NONE DEFINED

MAJOR: AQL 4.0%	Req't. Par.	QA. Par.	Test Method
101. Physical failure (post vibration)	3.6.1	4.8.4.2.1	Visual-Tactile Device, Special Testing (DST
102. Shock test	3.6.2	4.8.4.3	Visual-Tactile DST
103. Operating temperature test	3.6.5	4.8.4.5	Visual-Tactile DST

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TABLE III - CLASSIFICATION OF DEFECTS

Use Inspection Level L7, and Sampling Plan Table IV-A
of Standard MIL-STD-105

CRITICAL: NONE DEFINED

MAJOR: AQL 2.5%	Req't. Par.	QA. Par.	Test Method
101. Problem accuracy	3.7.2	4.8.2	DST
102. Backlash	3.7.3.1	4.8.3	DST

4.8.1 Problem accuracy. The "Computing Sight Settings" and "Simulated Gun Elevation" for each check point shown in Figure 2 shall be introduced into the sight. The line of sight of the telescope shall then fall on the respective check points of the test target, within the required tolerance. Accuracy readings shall be read using both the increasing and decreasing angle of approach. The "average error" is defined as one-half the sum of both radial readings. The backlash figure of merit" is defined as the sum of the computed average errors" divided by the number of check points at which readings are taken.

4.8.3 Backlash.

4.8.3.1 Vertical.- The speed shall be set at 700 miles per hour, angle of approach at plus 1600 mils, and climb and dive at zero. The telescope then shall be set on target, using hand pressure, and test shall be made as follows:

(a) Gently apply and remove a torque of 6 pound inches about the horizontal axis of the telescope holder and note the line of sight at the target.

(b) Reverse the direction of torque applied in (a) and note the line of sight. The difference between the readings at (a) and (b) above is vertical backlash.

4.8.3.2 Azimuth.- The speed shall be set at 0 miles per hour, angle of approach at zero 0 mils, and climb and dive at zero. The telescope shall be set on target, using hand pressure, and test shall be made as follows:

(a) Gently apply and remove a torque of 6 pound-inches about the vertical axis of the resolution mechanism and note the deflection of the line of sight at the target.

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(b) Reverse the direction of torque applied and note the deflection. The difference between the readings at. (a) and (b) above is computer and resolution mechanism backlash.

(c) Repeat the azimuth backlash test as specified In (a) and (b) above, with the computer set at 700 mph, angle of approach at plus 1600 mils, and climb and dive at zero.

4.8.4 Environmental.

4.8.4.1 General.- The environmental tests specified shall be limited to those obtained in this paragraph. Three of the first five sights and one out of every-additional 100 or fraction thereof shall be tested.

4.8.4.2 Vibration.- The computing sight shall be vibrated in accordance with Standard MIL-STD-202, Method 201, except the motion shall be applied for a period of 10 minutes in each of the 3 mutually perpendicular directions (total 30 minutes).

4.8.4.2.1 Physical failure, post vibration.- Subsequent to vibration, the computing sight shall be subjected to a visual and tactile examination fur damage. The unit then shall be subjected to all the tests specified.

4.8.4.3 Shock.- The computing sight shall be mounted rigidly, using its mounting provisions, to simulate field use. A shock of 20 g’s shall be imparted three times in each direction along each of the three mutually perpendicular directions, making a total of eighteen shocks. Each shock Impulse shall have a time duration of 11 plus or minus 1 mlliseconds; the maximum g value shall occur at approximately 5.5 milliseconds and shall be within plus or minus 10 percent. Filters used shall have cut-off frequencies selected to eliminate all harmonics above the fundamental, from the measured shock pulse.

4.8.4.4 Storage Temperature.- This test shall be conducted in accordance with Procedure II of MIL-F-13926 to the conditions specified. Upon completion of the specified exposure, the computing sight shall be inspected for damage, and then subjected tothe test specified in 4.8, and shall meet the requirements specified. This test shall be performed on the production model only.

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4.8.4.5 Operating temperature.- When thermal equilibrium of the computing sight is reached at plus 150 degrees Fahrenheit and minus 40 degrees Fahrenheit, the unit shall be subjected to the tests in 4.8 and shall meet the requirements specified.

4.8.4.6 Rain.- With dust cover Assembly F7666367 installed, the computing sight shall be tested In accordance with Procedure I of MIL-F-13926. With the dust cover removed the computing sight shall be tested in accordance with Procedure I of MIL-F-13926 with the exception that the simulated rainfall shall not exceed 1/4 plus or minus 1\8 inch per hour. The computing sight shall show no evidence of water penetrating as a result of this test. This test shall be performed on the production model only.

4.8.4.7 Humidity.- The computed sight to be tested shall be placed in the test chamber in its normal operating position and exposed to one complete humidity condensation cycle as specified. Distilled or demineralized water having a pH value between 6.5 and 7.5 at 77 degrees Fahreheit shall be used to obtain the desired humidity. The unit under test shall be placed no closer than 6 inches from the sides of the chamber, and shall not be subject to radiant heat. Upon completion of the humidity exposure, the unit shall show no evidence of damage. It shall then be subjected to the tests in 4.8, and shall meet the requirements speciified. This test shall be performed on the production model only.

5. PREPARATION FOR DELIVERY

5.1 Packaging and packing.- The computing sight shall be cleaned, preserved, packaged, packed, and marked in accordance with Specification MIL-S-14487. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use.- The computing sight is an on-carriage sighting device designed to provide an effective means of controlling fire of the 40-mm Dual Automatic Gun: M2A1.

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6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification
(b) Selection of an applicable level of preservation, packaging and packing
(c) Applicable stock number
(d) Provision for production model including date of delivery see 3.1)

6.3 Design information.- Information regarding specific design not covered or referenced by this specification may be obtained from the procuring activity upon the specific request of the contractor.

6.4 Definitions.

6.4.1 Special testing device (DST).- Special testing devices are designed to check function of an assembled unit.

6.4.2 Standard measuring equipment (SME).- Standard measuring equipment (SME) includes the common hand-type measuring devices which are usually stocked by commercial supply houses for ready supply (shelf items), and which are normally used by an inspector to perform dimensional inspection of items under procurement. This category also includes commercial testing equipment.

6.4.3 Tactile.- Skillful manipulation or sensing.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any invention that may in any way be related thereto.

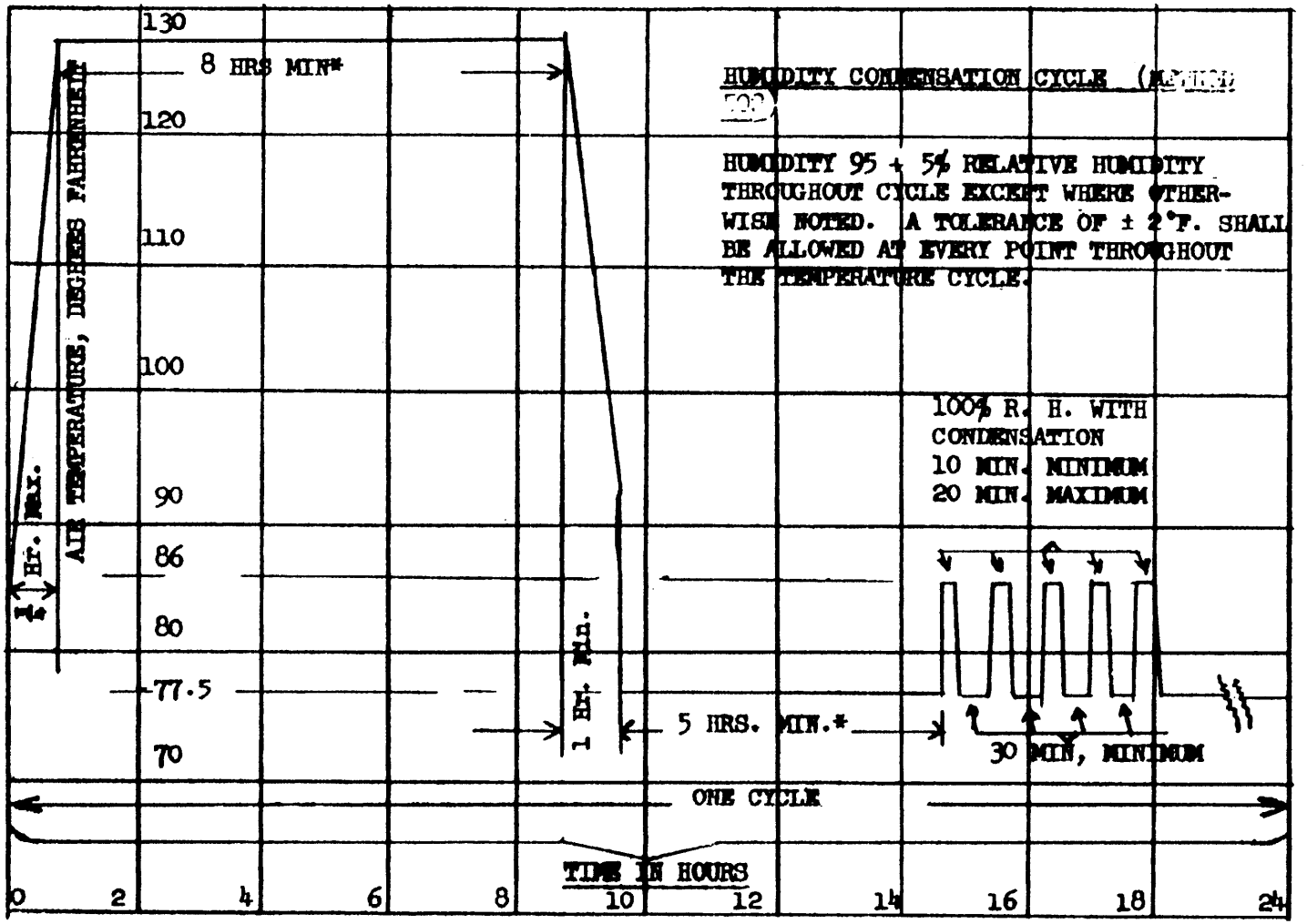
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* OR UNTIL EQUIPMENT STABILIZATION (WHICHEVER IS LONGER)

FIGURE I

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Simulated Gun Elevation	Computing Sight Settings			Test Target Layout Parameters			Check Points
	Dive Angle	Air Speed (MPH)	Approach Angle-Mils	L/b	V/b	Ev Inches	
		200	0	.1194	-.0090	0	1
		200	3200	-.1194	-.0090	0	2
Zero	Zero	400	0	.2388	-.0090	0	3
		400	3200	-.2388	-.0090	0	4
Mils	Degrees	700	0	.4179	-.0090	0	5
		700	800	.4194	-.0128	0	6
		700	2400	-.4194	-.0128	0	7
		700	3200	-.4179	-.0090	0	8
		360	0	.2149	-.0064	0	9
		360	1600	0	.1717	.08	10
800	Zero	360	3200	-.2149	-.0064	0	11
		360	-1600	0	-.1374	.06	12
Mils	Degrees	700	0	.4179	-.0064	0	13
		700	1600	0	.4104	.46	14
		700	3200	-.4179	-.0064	0	15
		700	-1600	0	-.2330	.17	16
		700	0	.2955	.2865	.24	17
Zero	45	700	800	.2641	.3621	.37	18
Mils	Degrees	700	2400	-.2641	.3621	.37	19
		700	3200	-.2955	.2865	.24	20

b = Distance from axis of rotation of telescope to test target
L = Lateral deflection V = Vertical deflection Ev = parallax correction

Note - The vertical coordinates of the test target should be corrected for telescope vertical parallax by adding algebraically the values of E_v to the values of V computed from the above tables. It is to be noted that the values of E_v are all positive and that this parallax correction Is Independent of the distance of the test target from the test fixture *on* which the sight Is mounted.

FIGURE 2

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